Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (Bsc.IT Semester III)

Data Structures Practical

Practical - IV

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Date of Assignment :	Date/Time of Submission :15/07/2026

A. Write a program to: a. Implement a queue using an array.

Code:

```
from array import array

queue = array('i') # Integer array

# Enqueue
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue:", queue)

# Dequeue
print("Deleted:", queue[0])
queue.pop(0)

print("Queue after Dequeue:", queue)

print("Kaushal Yadav 48A")
```

Output:

```
=== RESTART: C:/Users/IT-46/OneDrive/Desktop/kaushal yadav/practical4/ds4
Queue: array('i', [10, 20, 30])
Deleted: 10
Queue after Dequeue: array('i', [20, 30])
Kaushal Yadav 48
```

B.Simulate a simple queuing system (e.g., customer service queue).

Code:

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```
from array import array

# Create Queue
customers = array('i')

# Add Customers
customers.append(101)
customers.append(102)
customers.append(103)

print("Customers in Queue:", customers)

# Serve First Customer
print("Served Customer:", customers[0])
customers.pop(0)

print("Queue After Serving:", customers)

print("Kaushal Yadav 48")
```

Output:

```
= RESTART: C:/Users/IT-46/OneDrive/Desktop/kaushal yadav/practical4/Practical 4B
.py
Customers in Queue: array('i', [101, 102, 103])
Served Customer: 101
Queue After Serving: array('i', [102, 103])
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```